

SOFTWARE ENGINEERING

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS07	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

1. To familiarize with software engineering methods, practices and its applications.
2. To familiarize with software requirements and the SRS documents.
3. To facilitate students in software design.
4. To familiarize with testing strategies.
5. Understanding working knowledge, quality management and risk management of large software development projects.

COURSE OUTCOMES

1. Analyze software development life cycle and select appropriate model for software application.
2. Analyze the customer's requirements for a project to be developed and formulate the software requirements document.
3. Conceptualize the system through design on architectural modelling and user interface.
4. Classify software testing strategies and recommend testing techniques during the construction of software projects.
5. Analyze the risk management, quality management and quality standards before releasing the software product.

UNIT - I	INTRODUCTION TO SOFTWARE ENGINEERING	CLASSES: 10
Introduction to Software Engineering: The evolving role of software, changing nature of software, software myths. A Generic view of process: Software engineering- a layered technology, a process framework, the capability maturity model integration (CMMI). Process models: The waterfall model, Spiral model and Agile methodology		
UNIT - II	SOFTWARE REQUIREMENTS & REQUIREMENTS ENGINEERING PROCESS	CLASSES: 10
Software Requirements: Functional and non-functional requirements, user requirements, system requirements, interface specification, the software requirements document. Requirements engineering process: Feasibility studies, requirements elicitation and analysis, requirements validation, requirements management		
UNIT - III	DESIGN ENGINEERING	CLASSES: 10
Design Engineering: Design process and design quality, design concepts, the design model. Creating an architectural design: software architecture, data design, architectural styles and patterns, architectural design, conceptual model of UML, basic structural modeling, class diagrams, sequence diagrams, collaboration diagrams, use case diagrams, component diagrams.		

UNIT – IV	TESTING STRATEGIES	CLASSES: 10
Testing Strategies: A strategic approach to software testing, test strategies for conventional software, black-box and white-box testing, validation testing, system testing, the art of debugging. Metrics for Process and Products: Software measurement, metrics for software quality.		
UNIT - V	RISK MANAGEMENT & QUALITY MANAGEMENT	CLASSES: 10
Risk management: Reactive Vs proactive risk strategies, software risks, risk identification, risk projection, risk refinement, RMMM. Quality Management: Quality concepts, software quality assurance, software reviews, formal technical reviews, statistical software quality assurance, software reliability, the ISO 9000 quality standards.		
TEXT BOOKS		
1. Software Engineering, A practitioner's Approach- Roger S. Pressman, 6th edition, McGraw Hill International Edition. 2. Software Engineering- Sommerville, 7th edition, Pearson Education.		
REFERENCE BOOKS		
1. The unified modeling language user guide Grady Booch, James Rumbaugh, Ivar Jacobson, Pearson Education. 2. Software Engineering, an Engineering approach- James F. Peters, Witold Pedrycz, John Wiley. 3. Software Engineering principles and practice- Waman S Jawadekar, The McGraw-Hill Companies. 4. Fundamentals of object-oriented design using UML Meiler page-Jones: Pearson Education.		